

Multi-Agent Scientific Paper Review

Powered by Phi-3 Mini · May 03, 2026 18:06

File	Final_paper_kjns_210426 (1).pdf
Words	12,362
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	llama3.2

■ **AI-ASSISTED REVIEW NOTICE:** This report was generated by an artificial intelligence system using Phi-3 Mini as the language model. It is intended as a supplementary tool to assist human peer reviewers. All findings must be validated by a qualified domain expert before editorial decisions are made. The AI may miss context-specific nuances and should not be used as the sole basis for acceptance or rejection.

Editorial Decision

MAJOR REVISION

Weighted Score: 3.1 / 5.0
Confidence: 72%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	3.5	Good	■■■■■
Manuscript Structure	4.0	Good	■■■■■
Methodology & Equations	2.5	Weak	■■■■■
Statistical Analysis	2.8	Weak	■■■■■
Results & Evidence	3.0	Acceptable	■■■■■
Discussion & Conclusions	4.2	Good	■■■■■
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	2.5	Weak	■■■■■
References & Citations	2.0	Weak	■■■■■
Ethics & Transparency	3.5	Good	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 3.5 / 5.0 | Confidence: 80%

Major Comments:

- [TITLE] could be more concise (15 words max)

Recommendations:

- [ABSTRACT] Provide clear explanations for undefined abbreviations and references
-

StructureReviewer

Manuscript structure and logical coherence

Score: 4.0 / 5.0 | Confidence: 80%

Weaknesses:

- Introduction lacks clarity and is incomplete
- Transitions between sections could be improved

Major Comments:

- The manuscript could benefit from a clearer introduction that sets the stage for the research question.
- The discussion section feels somewhat disconnected from the rest of the manuscript.

Recommendations:

- Revise Introduction to improve clarity and completeness
 - Improve transitions between sections
-

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 2.5 / 5.0 | Confidence: 55%

Weaknesses:

- Lack of clarity in the presentation sequence and logical flow.
- The model produced a very low methodology score despite detected method, experiment, and validation sections; score was floored pending human review.

Major Comments:

- The authors need to reorganize their manuscript to follow a clear problem → model → method → experiment → results → conclusion sequence.

Recommendations:

- Reorganize the manuscript to improve logical flow and presentation sequence.
 - Provide more details on the dataset used for validation.
 - Review methodology manually against the detected article map before treating this score as final.
-

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: 2.8 / 5.0 | Confidence: 60%

Weaknesses:

- Insufficient justification for sample size (30 independent runs)

Major Comments:

- Kruskal-Wallis test may not be the most appropriate choice for this data type

Minor Comments:

- CV is interpreted correctly, but its relevance to the problem might be questioned

Recommendations:

- Consider using alternative tests (e.g., ANOVA) or more robust nonparametric alternatives
-

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **4.0 / 5.0** | Confidence: 80%

Weaknesses:

- No specific numerical values or data presented in tables

Major Comments:

- Figures are well-structured and clearly labeled.
- Equations could be more clearly defined with context.

Recommendations:

- Consider adding more detail to the table headers for clarity.
-

ResultsReviewer

Results validity and empirical support

Score: **3.0 / 5.0** | Confidence: 70%

Weaknesses:

- Lack of specific quantitative results, e.g., mean absolute error or correlation coefficient between predicted and measured values.

Major Comments:

- The Results section could benefit from more detailed explanations of the experimental setup and data preprocessing steps.
- It is unclear how the validation addresses all three key aspects (dynamic fidelity, parametric robustness, and multi-objective trade-off consistency).

Recommendations:

- Provide specific quantitative results to support the claims made in the Results section.
 - Consider adding more details about the experimental setup and data preprocessing steps.
-

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **4.2 / 5.0** | Confidence: 80%

Weaknesses:

- The Discussion could benefit from a more in-depth comparison with prior work, beyond just citing references.

Major Comments:

- The conclusions seem to be directly derived from the results, but it would be beneficial to discuss potential limitations and future directions more explicitly.

Recommendations:

- Consider adding a discussion on the implications of EANN's improved robustness and Pareto consistency in real-world applications.
-

WritingReviewer

Scientific writing quality, clarity, and style

Score: **2.5 / 5.0** | Confidence: 70%

Weaknesses:

- type: WRITING CRITERIA; score: 4
- type: WRITING CRITERIA; score: 3
- The model produced a very low writing score without parser evidence that the manuscript is unreadable; score was floored pending human review.

Major Comments:

- Avoid using 'This paper proposes a novel...' as an opener.

Recommendations:

- Define technical terms on first use.
 - Use connectors and transitions effectively.
 - Inspect representative paragraphs manually for grammar, terminology consistency, and transition quality.
-

ReferencesReviewer

References quality, recency, and relevance

Score: **2.0 / 5.0** | Confidence: 80%

Weaknesses:

- References section is empty except for a few references.
- Only one recent reference (2021) is present, while seminal works are missing

Major Comments:

- The manuscript lacks a comprehensive literature review and relies heavily on outdated references.

Recommendations:

- Provide the full reference, including title, journal name, volume number, and page numbers
-

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **3.5 / 5.0** | Confidence: 60%

Strengths:

- No clear human/animal-subject signal was detected in the parsed manuscript.

Weaknesses:

- Conflict-of-interest declaration was not detected.
- The model response was not valid JSON, so a deterministic ethics/transparency fallback was used.

Major Comments:

- Conflict-of-interest declaration was not detected.
- The model response was not valid JSON, so a deterministic ethics/transparency fallback was used.

Recommendations:

- Include a conflict-of-interest declaration.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] [TITLE] could be more concise (15 words max)
- [2] The manuscript could benefit from a clearer introduction that sets the stage for the research question.
- [3] The discussion section feels somewhat disconnected from the rest of the manuscript.
- [4] The authors need to reorganize their manuscript to follow a clear problem → model → method → experiment → results → conclusion sequence.
- [5] Kruskal-Wallis test may not be the most appropriate choice for this data type
- [6] Figures are well-structured and clearly labeled.
- [7] Equations could be more clearly defined with context.
- [8] The Results section could benefit from more detailed explanations of the experimental setup and data preprocessing steps.
- [9] It is unclear how the validation addresses all three key aspects (dynamic fidelity, parametric robustness, and multi-objective trade-off consistency).
- [10] The conclusions seem to be directly derived from the results, but it would be beneficial to discuss potential limitations and future directions more explicitly.
- [11] Avoid using 'This paper proposes a novel...' as an opener.
- [12] The manuscript lacks a comprehensive literature review and relies heavily on outdated references.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] CV is interpreted correctly, but its relevance to the problem might be questioned

Specific Improvement Recommendations

1. [ABSTRACT] Provide clear explanations for undefined abbreviations and references
2. Revise Introduction to improve clarity and completeness
3. Improve transitions between sections
4. Reorganize the manuscript to improve logical flow and presentation sequence.
5. Provide more details on the dataset used for validation.
6. Review methodology manually against the detected article map before treating this score as final.
7. Consider using alternative tests (e.g., ANOVA) or more robust nonparametric alternatives
8. Consider adding more detail to the table headers for clarity.
9. Provide specific quantitative results to support the claims made in the Results section.
10. Consider adding more details about the experimental setup and data preprocessing steps.
11. Consider adding a discussion on the implications of EANN's improved robustness and Pareto consistency in real-world applications.
12. Define technical terms on first use.